Cleaning robot: proposal and route study

ATOMQUEST 2025 | Team Super Kings | Gajendra, 2301ME18

Abstract - short summary

This report explains Team Super Kings' proposed cleaning robot and a later grid-based software study. The team was Harsh Raj, Rakesh Hembram and Gajendra. The proposal combines brush, vacuum and mop modules. The runnable program explores obstacle-aware route coverage; it does not demonstrate a constructed robot or cleaning performance.

Purpose and background

The proposed system aimed to clean a floor in quadrants using onboard sensing and locomotion. The diagram separates the proposed hardware idea from the Python route simulation documented here.

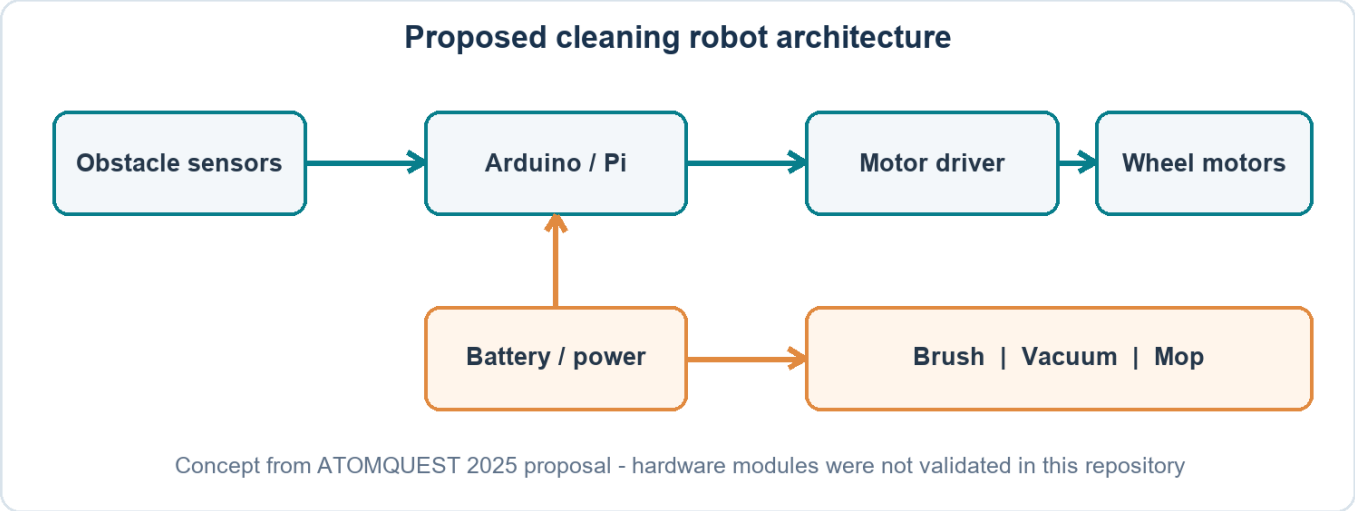


Figure 1. Concept diagram derived from the team proposal; component selection and integration remain unvalidated.

Methods - how the study was carried out

- Represent floor cells as open (.) or blocked (#); start at the first open cell in the default map.
- Use breadth-first search to find a shortest obstacle-free route to the nearest uncleaned reachable cell.
- Repeat until every cell connected to the start has been visited; count moves and report unreachable cells separately.

Cleaning robot: proposal and route study | Work and results

What was done

The items below describe the work represented by this repository and the course context.

• The team prepared the ATOMQUEST proposal; the original proposal PDF remains in this project folder.

• A Python simulator and tests were prepared for this portfolio to demonstrate grid coverage.

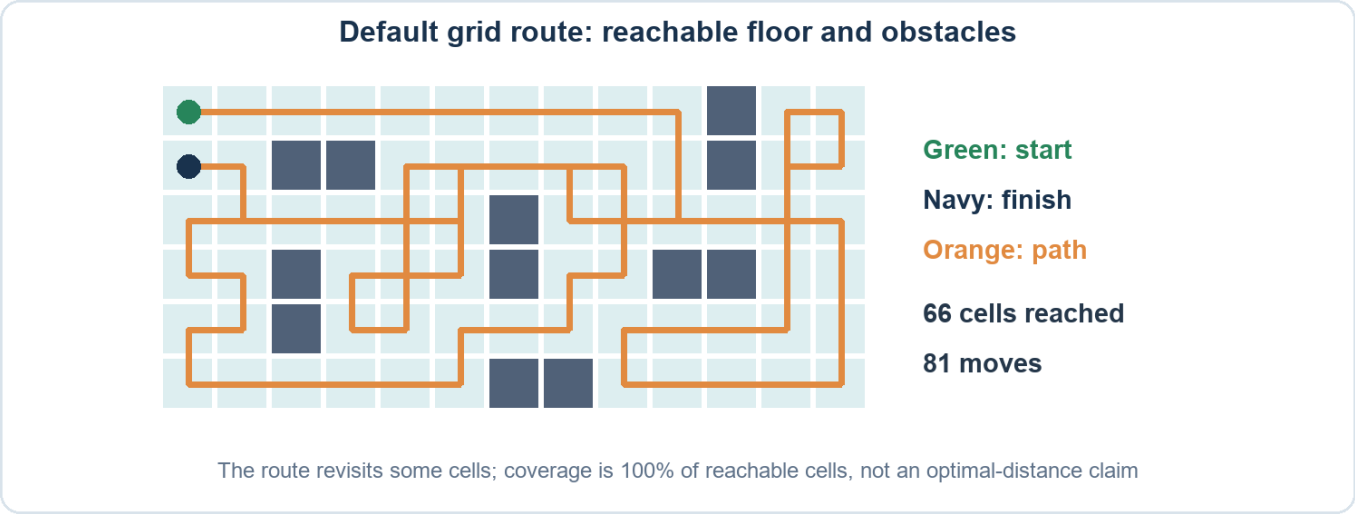


Figure 2. Generated route on the default map. The orange line shows travel, not a physical robot trace.

Result and interpretation

On the default 13 by 6 map, the software reaches 66 of 66 open cells in 81 moves. Coverage of reachable floor is 100%. The route may revisit cells, so this is a coverage result and not a shortest-overall-route result.

Limits and next validation step

The map assumes known static obstacles and four-direction movement. No localization, wheel slip, dynamic obstacle, battery, suction, mopping or hardware measurements are modeled. A physical build would require wiring checks, sensor calibration, collision tests and measured coverage.

Reproduce it

Run: python projects/cleaning_bot/simulator.py

Open project code on GitHub